QUANTITATIVE RESEARCH ASSIGNMENT

Submission Report

Bank Nifty 09:20 Short Strangle Backtest

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Bank Nifty Weekly Options | Short Strangle | Jan 2023 to Jan 2024

1. Executive Summary

This report documents the submission for the Qode Quant Research Analyst technical assignment: a vectorized backtest of a 09:20 AM Bank Nifty short strangle system over 247 trading days (494 option legs). The strategy sells one call and one put at strikes nearest to Rs. 50 premium, with a 50% stop-loss per leg and scheduled exit at 15:20. All deliverables requested by Qode are included: Python code, Excel workbook, charts, and runtime profiling.

Metric	Value
Compounded Annual Growth Rate (CAGR)	8.27%
Maximum Drawdown	-3.11%
Final NAV (Base 100)	108.2593
Total Gross P&L	Rs. 8259.34
Total Trades	494 (247 days x 2 legs)
Win Rate (Combined)	49.60%
Total Runtime	39.72 seconds

2. Assignment Objective

Develop a backtest for a 09:20 AM short strangle trading system using one year of Bank Nifty index and options data. The system must demonstrate options strategy design, vectorized data manipulation, backtesting accuracy, statistical analysis, and computational efficiency (target: under 60 seconds end-to-end).

2.1 Core Requirements

- Strike selection: CE and PE with 09:20 1-minute close nearest to Rs. 50
- Entry at 09:20; exit at 15:20 or 50% stop-loss per leg (checked via High column)
- Fixed position size: 1 lot x 15 quantity, no compounding
- Trade week-1 weekly options (Wednesday expiry); trade every trading day
- Vectorized implementation with no per-day Python loops
- Excel output with Guide, Tradesheet, and Statistics worksheets

3. Strategy Overview

A **short strangle** is a neutral options strategy that profits from time decay when the underlying remains range-bound. The trader simultaneously sells an out-of-the-money call (CE) and put (PE) with the same expiration. Premium is collected at entry; the position is closed by buying back the options at a lower price (profit) or higher price (loss).

3.1 Trade Lifecycle

Phase	Time	Action
Strike Selection	09:20:59	Identify CE and PE with Close nearest to Rs. 50
Entry	09:20:59	Sell both legs at 1-min Close (short)
Monitoring	09:21 - 15:20	Check High each minute for 50% SL breach
Stop-Loss Exit	First breach	Buy back at Entry x 1.5
Scheduled Exit	15:20:59	Buy back at 1-min Close if SL not hit

4. Methodology

4.1 Strike Selection Module

At 09:20:59 each trading day, all available CE and PE contracts are evaluated independently. The contract whose 1-minute Close price minimizes |Close - 50| is selected. Tie-breaking: higher strike for CE, lower strike for PE. Implementation uses vectorized groupby on approximately 120 contracts per day.

4.2 Signal Generation and Stop-Loss

For each selected leg, a 50% stop-loss is defined as SL Price = Entry Price x 1.5. Since the position is short, loss occurs when the option price rises. Each 1-minute bar from 09:21:59 through 15:20:59 is scanned; if High is greater than or equal to SL Price, the leg exits at SL Price at the first breach minute. Otherwise, exit occurs at 15:20:59 Close.

4.3 Position Sizing

Fixed quantity of 15 units (1 lot x lot size 15) per leg per day. No compounding. Available capital tracks cumulative P&L for reporting only (starting capital: Rs. 1,00,000).

4.4 P&L Calculation

Gross P&L = Entry Value - Exit Value = (Entry Price - Exit Price) x Quantity. NAV starts at base 100 and updates daily: NAV = 100 + (Cumulative P&L / Starting Capital) x 100.

5. Implementation Architecture

Module	File	Function
Data Loading	short_strangle_backtest.py	load_options(), load_spot()
Strike Selection	short_strangle_backtest.py	select_strikes()
Signal and SL	short_strangle_backtest.py	apply_signals()
Trade Sheet	short_strangle_backtest.py	build_tradesheet()
Statistics	short_strangle_backtest.py	compute_statistics()
Export	short_strangle_backtest.py	export_excel(), save_plots()
Interactive	short_strangle_backtest.ipynb	Step-by-step notebook

5.1 Key Assumptions

- Dataset contains week-1 (nearest Wednesday expiry) Bank Nifty weekly options only
- Wednesday is expiry day; all trading days included including expiry Wednesdays
- SL fill at exactly 1.5x entry when High breaches; no slippage or transaction costs
- Duplicate Date/Ticker/Time rows deduplicated (keep last)
- Spot underlying merged on minute key where available

6. Performance Results

6.1 Summary Metrics

Metric	Value
Starting Capital (INR)	100000
Base NAV	100.0
Final NAV	108.2593
Total Gross P&L (INR)	8259.34
CAGR	8.27%
Max Drawdown	-3.11%
Total Trading Days	247

Metric	Value
Total Trades	494
SL Exits	0
Scheduled Exits	494

6.2 Win / Loss Analysis

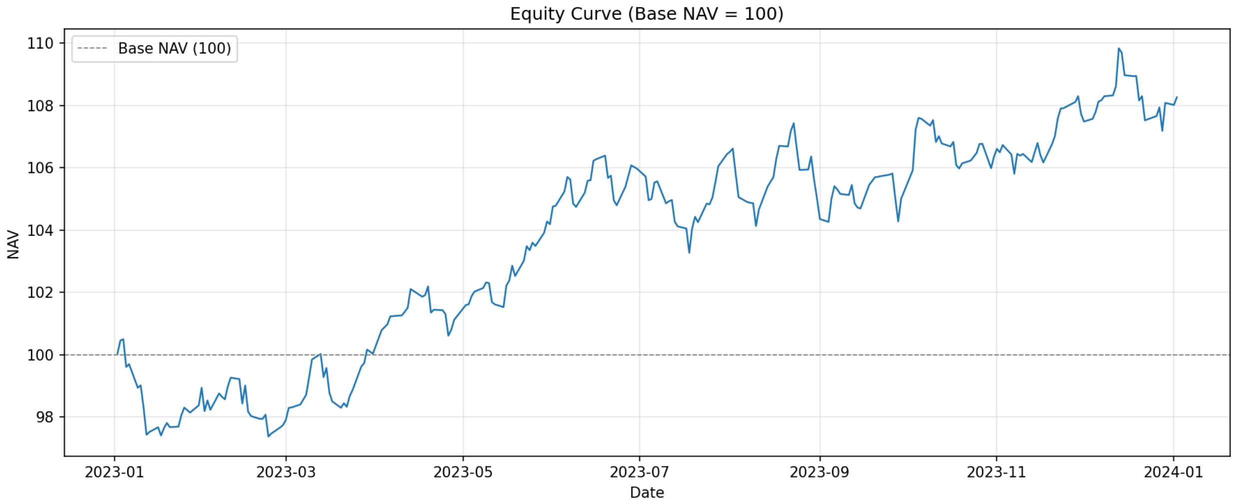
Category	Winners	Losers	Win %	Loss %	Avg % P&L
CE	123	124	49.80%	50.20%	2.13%
PE	122	125	49.39%	50.61%	0.94%
Combined	245	249	49.60%	50.40%	1.53%

6.3 Average % P&L: Expiry vs Non-Expiry Days

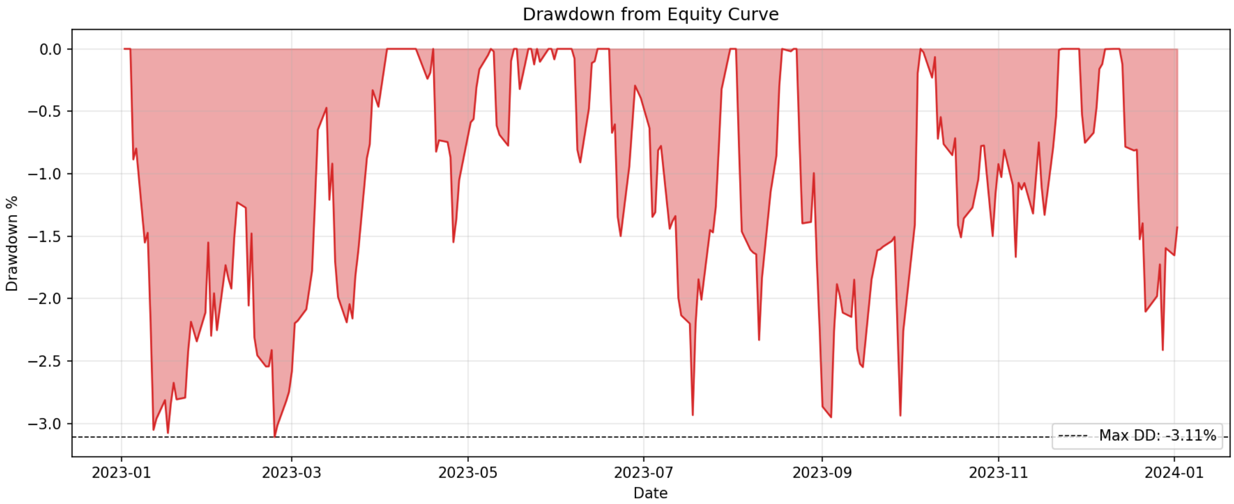
Segment	Day Type	Trades	Avg % P&L	Avg Gross P&L (Rs.)
CE	Expiry Days (Wed)	52	11.23%	87.08
CE	Non-Expiry Days	195	-0.30%	6.68
PE	Expiry Days (Wed)	52	-0.89%	2.01
PE	Non-Expiry Days	195	1.43%	11.92
Combined	Expiry Days (Wed)	104	5.17%	44.55
Combined	Non-Expiry Days	390	0.57%	9.30

7. Equity Curve and Drawdown

7.1 Equity Curve (Base NAV = 100)



7.2 Drawdown Profile



7.3 Equity Curve Table (First and Last 10 Days)

Date	Daily P&L	NAV	Drawdown %
2023-01-02	36.379999999999995	100.0364	0.0
2023-01-03	412.5	100.4489	0.0
2023-01-04	42.370000000000005	100.4912	0.0
2023-01-05	-889.88	99.6014	-0.8855

Date	Daily P&L	NAV	Drawdown %
2023-01-06	88.5	99.6899	-0.7975
2023-01-09	-757.5	98.9324	-1.5513
2023-01-10	78.38	99.0108	-1.4733
2023-01-11	-723.37	98.2874	-2.1931
2023-01-12	-862.88	97.4245	-3.0518
2023-01-13	89.25	97.5138	-2.9629
2023-12-19	7.5	108.9437	-0.8072
2023-12-20	-788.25	108.1555	-1.5249
2023-12-21	141.0	108.2965	-1.3965
2023-12-22	-777.75	107.5187	-2.1046
2023-12-26	136.5	107.6552	-1.9803
2023-12-27	279.75	107.935	-1.7256
2023-12-28	-754.13	107.1808	-2.4123
2023-12-29	897.0	108.0778	-1.5955
2024-01-01	-64.12	108.0137	-1.6539
2024-01-02	245.62	108.2593	-1.4303

7.4 Monthly % P&L (from NAV)

Month	End NAV	Monthly % P&L
2023-01	98.9331	-1.0669
2023-02	97.7312	-1.2149
2023-03	100.0255	2.3475
2023-04	101.1134	1.0876
2023-05	104.1827	3.0356
2023-06	105.9666	1.7123

Month	End NAV	Monthly % P&L
2023-07	106.4335	0.4406
2023-08	105.0186	-1.3293
2023-09	104.9999	-0.0179
2023-10	106.3514	1.2872
2023-11	107.7231	1.2898
2023-12	108.0778	0.3293
2024-01	108.2593	0.1679

8. Runtime Performance Analysis

The backtest completes in **39.72 seconds**, within the 60-second target specified in the assignment.

Step	Time (sec)
Data Loading (Options)	31.802
Data Loading (Spot Index)	0.970
Strike Selection	0.063
Backtest / Signal and Stop-Loss	5.050
Trade Sheet Construction	0.065
Statistical Analysis	0.014
Chart Generation	0.656
Excel Export	0.001
Total Runtime	**39.715**

8.1 Optimization Techniques

- Selective column loading and categorical dtypes for Ticker and Call/Put
- Early time-window filter (09:20 to 15:20) reducing dataset size
- Vectorized groupby for strike selection and stop-loss detection
- Single-pass CSV load with duplicate row deduplication

9. Submission Deliverables

Deliverable	File	Status
Python Backtest Code	short_strangle_backtest.py	Complete
Jupyter Notebook	short_strangle_backtest.ipynb	Complete
Excel - Guide Sheet	backtest_output.xlsx (Sheet 1)	Complete
Excel - Tradesheet	backtest_output.xlsx (Sheet 2)	Complete
Excel - Statistics	backtest_output.xlsx (Sheet 3)	Complete

Deliverable	File	Status
Equity Curve Chart	equity_curve.png	Complete
Drawdown Chart	drawdown.png	Complete
Runtime Profiling	Guide sheet and this report	Complete
Dependencies	requirements.txt	Complete

9.1 Tradesheet Columns

Column	Included
Entry Date	Yes
Entry Time	Yes
Exit Date	Yes
Exit Time	Yes
Option Ticker	Yes
Strike Price	Yes
Option Type	Yes
Entry Price	Yes
Exit Price	Yes
Quantity	Yes
Entry Value	Yes
Exit Value	Yes
Gross P&L	Yes
Cumulative P&L	Yes
Available Capital	Yes
Banknifty Underlying Close	Yes
Exit Reason	Yes
Is Expiry Day	Yes

Column	Included
% P&L	Yes

10. Appendix: Sample Trades

First five trades from the Tradesheet demonstrating entry, exit, P&L, and capital tracking.

Entry Date	Option Type	Option Ticker	Entry Price	Exit Price	Gross P&L	Exit Time
2023-01-02	PE	BANKNIFTY42000PE	46.8	19.35	411.75	15:20:59
2023-01-02	CE	BANKNIFTY44000CE	50.05	75.07	-375.37	09:44:59
2023-01-03	PE	BANKNIFTY42500PE	46.5	20.45	390.75	15:20:59
2023-01-03	CE	BANKNIFTY44000CE	45.15	43.7	21.75	15:20:59
2023-01-04	PE	BANKNIFTY43000PE	54.85	82.28	-411.38	09:43:59

10.1 Interpretation Guide

Winners (short options): Exit Price is less than Entry Price. **Losers:** Exit Price is greater than Entry Price, typically from stop-loss events. **Expiry days (Wednesdays):** accelerated time decay can affect CE/PE performance. **NAV:** Normalized portfolio value starting at 100. **Max Drawdown:** Largest peak-to-trough decline in NAV.

Report generated from live backtest output on 14 June 2026. All figures reconcile with backtest_output.xlsx.